

CSE-2422 Computer Algorithms Lab

List of Lab Experiments

Students are expected to implement the following algorithms for getting familiar to those.

A. Implementation of sorting algorithms

1. Implementation of Insertion-sort
2. Implementation of Merge-sort
3. Implementation of Quick-sort with its randomized version
4. Implementation of Heap and Heap-sort
5. Implementation of Counting sort
6. Implementation of Radix sort
7. Implementation of Priority queue using binary heap

B. Implementation of dynamic programming algorithms

8. Solving matrix-chain multiplication problem
9. Solving longest common subsequence problem

C. Implementation of greedy algorithms

10. Solving activity selection problem
11. Implement Huffman tree and generating prefix code
12. Solving Fractional Knapsack problem

D. Implementation of string-matching algorithms

13. Implementation of Naive string-matching algorithm
14. Implementation of Rabin-Karp algorithm
15. Implementation of Knuth-Morris-Prat algorithm

E. Implementation of basic graph algorithms

16. Implementation of Breadth First Search
17. Implementation of Depth First Search

F. Implementation of minimum spanning tree algorithms

18. Implementation of Kruskal's Algorithm for finding minimum spanning tree
19. Implementation of Prim's Algorithm for finding minimum spanning tree

G. Implementation of shortest path algorithms

20. Implementation of Dijkstra's algorithm for solving single-source shortest path problem

- 21. Implementation of Bellman-Ford's algorithm for solving single-source shortest path problem
- 22. Implementation of Floyd-Warshall's algorithm for solving all-pairs shortest path problem

H. Implementation of computational geometry algorithms

- 23. Algorithmically Determine if two line segments intersect
- 24. Finding the convex hull of a set of points using Graham's scan algorithm

I. Implementation of backtracking and number theoretic algorithms

- 25. Implementation of extended Euclid's algorithm for finding GCD
- 26. Implementation of different prime number generation algorithms
- 27. Finding the value of $X^n \bmod m$ through modular exponentiation
- 28. Solving N-Queen Problem
- 29. Generate permutation through backtracking